

**AI-BASED CANDIDATE RECOMMENDATION SYSTEM
FOR COMPANY VACANCIES: EMPHASIZING SOFT
SKILLS AND COMPETENCIES**

R25-040

Research Group Final Report

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Sri Lanka Institute of Information Technology
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August 2025

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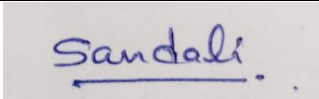
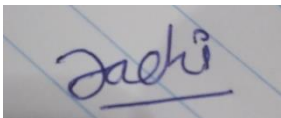
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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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Working on this final year research project has been an immensely enriching academic experience for us. Through the process of developing our research proposal, we have gained valuable insights into the research process, including how to formulate research questions, design suitable methodologies and anticipate potential challenges. We would like to express our heartfelt gratitude to our supervisor, Prof. Samantha Thelijjagoda for his invaluable guidance, feedback and support throughout this project. His constant input and insightful suggestions are greatly appreciated. We would also like to express our gratitude to our co- supervisor, Ms. Poorna Panduwawala, for her unwavering guidance, feedback, and support throughout the project. We owe a debt of gratitude to all the lecturers at SLIIT as well. We look forward to continuing to develop our research skills and contribute meaningful insights through this project

ABSTRACT

Recruitment and candidate evaluation have, for decades, been carried out primarily through résumés, academic qualifications and assessments of technical expertise. While these approaches remain useful for identifying a candidate's formal competencies, they provide only a partial picture of overall suitability for a role. Modern workplaces demand much more than technical proficiency; they require strong soft skills and behavioral competencies such as confidence, stress management, communication and adaptability. Unfortunately, these crucial attributes are often overlooked or only superficially assessed during traditional interviews, where evaluations are highly dependent on subjective judgments made by recruiters.

This research addresses this gap by introducing an AI-based candidate recommendation system that places emphasis on soft skills, with a particular focus on confidence and stress-handling abilities. The system adopts a multi-modal analysis framework, incorporating four specialized modules designed to capture different behavioral cues. The first module evaluates speech and tone to identify whether a candidate communicates with confidence or hesitancy. The second module examines hand gestures, as non-verbal communication often reflects confidence. The third module analyzes eye-contact patterns, since maintaining appropriate gaze is a strong indicator of confidence, attentiveness and self-assurance during interviews. Finally, the fourth module monitors facial expressions to determine whether a candidate is stressed or composed under interview conditions. Each module produces a binary classification Confident or Non-Confident in the case of the first three modules, and Stress or Non-Stress for the facial expression module. These outputs are then aggregated to create a holistic assessment of the candidate's soft skills. A minimal decision-making skill component is also included to refine the overall evaluation. By combining these results, the system provides recruiters with a more objective and reliable recommendation, reducing human bias and subjectivity in candidate evaluation. The outcomes of this study demonstrate that an AI-driven multi-modal framework can significantly improve the fairness and depth of recruitment processes. The final integrated system, deployed as a web application, allows companies to evaluate candidates beyond their CVs and technical tests, thereby enabling more informed, fair and effective hiring decisions.

Keywords: Artificial Intelligence, Candidate Recommendation System, Soft Skills, Confidence Prediction, Speech and Tone Analysis, Hand Gesture Recognition, Eye Contact Detection, Facial Expression Analysis, Stress Detection, Recruitment Automation, Human Resource Management, Multi-Modal Analysis, Machine Learning, Web Application.

Table of Contents

DECLARATION	3
ACKNOWLEDGEMENT	4
ABSTRACT	4
Table of Contents	5
List of Figures	6
List of Tables	6
1. INTRODUCTION	7
1.1. Background Literature	9
1.2. Research Gap	11
1.3. Research Problem	13
1.4 Objectives	15
1.4.1 Main Objective	15
1.4.2. Specific Objectives	15
2. METHODOLOGY	19
2.1. Methodology	19
2.1.1 System Diagram	30
2.2 Commercialization Aspects of InsightHire	34
2.3 Implementation and Testing	38
2.3.1.1 Data Acquisition and Preprocessing	38
2.3.1.2 Machine Learning Models and Decision Logic	39
2.3.1.3 Web Application and User Interface	39
2.3.1.4 System Integration and Deployment	40
2.3.2 Testing and Validation:	41
2.3.2.1 System Optimization:	41
2.3.2.2 User Feedback and Final Revisions	42
2.3.2.3 Test Cases	44
3. RESULTS AND DISCUSSION	47
3.1. Results	47
3.2. Research Findings	49
3.3. Discussion	52
SUMMARY OF EACH STUDENT'S CONTRIBUTION	54
CONCLUSION	56
REFERENCES	58

List of Figures

Figure 1: Model Training Process to predict confidence through speech and tone analysis	28
Figure 2: Model Training Process to predict stress level through facial expressions	28
Figure 3: Model Training Process to predict confidence through hand gestures	28
Figure 4: Model Training Process to predict confidence through eye contact	29
Figure 5: System Overview Diagram.....	30
Figure 6: Gannt chart	60
Figure 7: Turnitin report	60

List of Tables

Table 1: Test case 1: Detecting Confidence Level via Voice Tone	42
Table 2: Test case 2: Detecting Stress through Facial Expressions	42
Table 3: Test case 3: Positive Hand Gesture Detection	43
Table 4: Test case 4: Engagement Detection by Eye Tracking.....	43
Table 5: Test case 5: System Performance Under Low-Light Conditions	44
Table 6: Test case 6: Multi-Modal Data Fusion Accuracy	44

1.INTRODUCTION

Recruitment and candidate evaluation have historically focused on resumes, academic qualifications and technical skills. While these criteria remain important, modern organizations increasingly recognize that technical expertise alone does not guarantee success in the workplace. Soft skills such as confidence, communication, adaptability and stress management play a vital role in determining how well an employee integrates into a team and performs under real-world conditions. Employers place significant emphasis on these behavioral attributes, yet traditional selection methods often fail to measure them adequately, resulting in hiring decisions that may not fully align with organizational needs ([6]; [7]; [26]; [17]; [40]; [54]).

Artificial Intelligence (AI) offers a transformative approach to address these limitations by providing objective, consistent and data-driven insights into candidate behavior. With advancements in computer vision, natural language processing and multimodal machine learning, researchers have developed systems capable of analyzing human communication beyond verbal responses, incorporating tone of voice, hand gestures, eye movements, and facial expressions ([31]; [44]; [51]; [42]; [58]; [61]). These technologies enable evaluation of not only what a candidate says but also how they present themselves, capturing subtle behavioral cues that reflect confidence and stress-handling abilities.

This research presents an AI-based candidate recommendation system that leverages multiple modalities to assess soft skills and behavioral competencies during interviews. The system integrates four core components. The first predicts confidence through speech and tone analysis, as vocal variations in pitch, intonation and rhythm often indicate levels of self-assurance or insecurity ([34]; [36]; [43]). The second component evaluates confidence using hand gesture recognition, as purposeful and open gestures are associated with assertiveness, while restricted or nervous movements may indicate hesitation ([47]; [48]; [49]). The third component assesses confidence through eye-contact detection, given that sustained gaze reflects confidence, engagement and self-assurance, whereas avoidance may suggest discomfort or low confidence ([1]; [2]; [5]). The fourth component predicts stress levels through facial expression analysis, drawing on research in affective computing that shows micro-expressions, facial tension and subtle muscle movements can reliably reveal stress ([11]; [12]; [19]; [25]).

Each component produces a binary outcome confident or non-confident for the first three modules, and stress or non-stress for the facial expression module. These outcomes are then aggregated into a unified framework to provide a holistic assessment of a candidate's behavioral profile. While the system primarily emphasizes confidence and stress-handling, decision-making skills are incorporated minimally to further refine candidate evaluation.

The outcomes of this research are integrated into a web-based application designed to support

recruiters in making data-driven hiring decisions. By consolidating insights from multiple behavioral modalities, the system enables companies to evaluate candidates beyond conventional CV-based screening and subjective interview impressions ([52]; [55]; [59]). This approach enhances fairness in recruitment and acknowledges that soft skills and emotional intelligence are critical predictors of professional success ([6]; [7]; [27]; [28]).

Finally, this study considers potential challenges including cultural variations in non-verbal communication, biases in AI datasets and ethical concerns regarding privacy and transparency ([40]; [45]; [46]; [57]; [58]). Addressing these issues is essential to ensure the responsible deployment of AI-based recruitment tools in real-world scenarios. Overall, this research aims to demonstrate that a multimodal AI-driven approach can provide a more comprehensive, accurate and fair assessment of candidates, improving both the efficiency and effectiveness of modern recruitment processes.

1.1. Background Literature

The role of soft skills in modern recruitment has become increasingly important, as organizations recognize that technical proficiency alone does not guarantee workplace success. Competencies such as confidence, stress management, communication and emotional intelligence significantly influence an employee's performance, teamwork and adaptability ([6]; [7]; [27]; [28]; [32]). Traditional recruitment methods, including resume screening and technical assessments, often fail to capture these behavioral traits, leading to incomplete evaluations of candidate potential ([17]; [40]; [54]).

Consequently, there has been growing interest in leveraging artificial intelligence to assess soft skills through objective, data-driven approaches.

Research in affective computing and human behavior analysis provides a foundation for AI-driven evaluation of candidate behavior. Emotion recognition using facial expressions, vocal cues and body gestures has been extensively studied in educational and professional contexts to measure engagement, confidence and stress ([1]; [2]; [3]; [4]; [19]; [20]). Facial expressions have been particularly emphasized as reliable indicators of underlying psychological states. Classical studies by Darwin ([11]) and subsequent work by Ekman and Friesen ([12]; [13]) demonstrate that universal facial expressions, micro-expressions, and muscle activations convey emotional information across cultural contexts. Modern research extends this understanding through deep learning approaches that automate facial emotion recognition, allowing systems to classify expressions in real-time and at scale ([20]; [21]; [25]). Such methods provide a basis for detecting stress and confidence during high-stakes situations such as interviews.

Speech and tone have also been widely investigated as markers of confidence and emotional state. Variations in pitch, intensity, rhythm, and prosody carry significant information about a speaker's confidence level ([34]; [35]; [36]; [37]; [38]; [39]; [43]). Computational models, including convolutional and recurrent neural networks, have been employed to analyze these vocal features, predicting emotional states and confidence with high accuracy ([37]; [38]; [43]). Open-source tools such as openSMILE enable efficient extraction of acoustic and prosodic features, facilitating the development of AI-based recruitment tools ([36]). Additionally, speech-based analysis has been shown to reduce human biases in evaluation, offering more equitable assessment of candidate behavior ([40]; [41]; [54]).

Non-verbal communication through hand gestures is another critical indicator of confidence. Research indicates that candidates exhibiting purposeful, open gestures are perceived as more confident and persuasive, whereas restrained or repetitive gestures may signal nervousness ([47]; [48]; [49]). Recent advances in computer vision and pose estimation, including frameworks such as MediaPipe, enable real-time hand gesture recognition ([50]). These systems can classify gestures automatically,

integrating the findings with other behavioral cues for comprehensive candidate evaluation ([51]; [58]).

Eye contact serves as a direct indicator of confidence, engagement and self-assurance. Studies in both human-robot interaction and educational contexts have demonstrated that sustained gaze is associated with higher levels of confidence and attention, whereas avoidance or frequent shifting of gaze often correlates with discomfort or low confidence ([1]; [2]; [5]). AI-based gaze tracking uses video analysis to measure eye movements accurately, providing quantifiable metrics that complement speech and gesture analysis.

Stress detection through facial expressions represents another critical component of soft skill evaluation. Stress manifests in subtle facial micro-expressions, brow tension and mouth movements, which can be captured and analyzed using computer vision and deep learning techniques ([11]; [12]; [14]; [19]; [20]; [21]; [25]). Recognizing stress in candidates allows organizations to assess how individuals manage pressure, an essential predictor of workplace resilience. The integration of multimodal signals including speech, gestures, eye contact, and facial expressions enables a richer understanding of candidate behavior than any single modality alone ([31]; [44]; [51]).

Ethical considerations are central to the adoption of AI in recruitment. Studies highlight the importance of transparency, fairness and privacy when using behavioral analysis tools ([30]; [45]; [46]; [57]; [61]). Bias in AI models, cultural differences in non-verbal communication, and the need for explainable decision-making are ongoing challenges that must be addressed to ensure equitable outcomes ([42]; [58]). Furthermore, practical deployment considerations such as real-time processing, system integration, and usability influence the effectiveness of AI-based recruitment systems ([52]; [53]; [55]; [60]).

In summary, prior literature underscores the feasibility and value of multimodal AI systems for evaluating soft skills and behavioral competencies. Research on speech, gestures, eye contact and facial expressions demonstrates that these modalities provide reliable indicators of confidence and stress. Combining these insights into a unified AI framework allows organizations to move beyond traditional recruitment practices, supporting fairer, more comprehensive, and data-driven hiring decisions ([6]; [7]; [26]; [31]; [44]; [52]; [59]). The proposed research builds upon these foundations to develop an integrated system capable of assessing candidates' confidence and stress-handling abilities across multiple modalities, contributing to more effective talent acquisition.

1.2. Research Gap

Despite the growing body of research on AI-driven behavioral analysis, several gaps remain in applying these methods specifically to recruitment and candidate evaluation. While prior studies have explored emotion recognition, confidence estimation in educational and human-computer interaction contexts ([1]; [2]; [3]; [4]; [5]; [18]; [19]; [20]), few have systematically integrated multiple behavioral modalities to assess candidates' soft skills in real-world hiring scenarios. Most existing systems focus on single modalities, such as facial expressions, speech, or gestures, which limits the reliability and comprehensiveness of candidate assessment ([31]; [44]; [51]).

Speech-based confidence analysis has been extensively studied, with numerous deep learning models and prosodic feature extraction techniques demonstrating high accuracy in controlled environments ([34]; [36]; [37]; [38]; [39]; [43]). However, these approaches often neglect the integration of complementary non-verbal cues, such as hand gestures or eye contact, which are essential for evaluating holistic confidence in an interview setting. Similarly, while hand gesture recognition and eye-contact analysis have shown promise for engagement, confidence detection ([47]; [48]; [49]; [50]; [58]), research rarely considers how these signals can be fused with vocal and facial data to create a comprehensive, multimodal assessment.

Stress detection through facial expressions has also been widely investigated ([11]; [12]; [14]; [19]; [20]; [21]; [25]). Existing systems can reliably detect stress or discomfort, but few studies have explored the integration of stress evaluation with confidence prediction to provide recruiters with actionable insights about candidates' behavioral resilience under pressure. Furthermore, current research often overlooks the practical implementation of such multimodal systems in real-world recruitment pipelines, including web-based platforms or tools that can provide interpretable and actionable recommendations to human decision-makers ([52]; [53]; [55]; [59]).

Another critical gap involves the ethical and cultural dimensions of AI-based evaluation. Biases in training datasets, differences in cultural norms for gestures, facial expressions and eye behavior and concerns regarding privacy and transparency remain under-addressed ([40]; [42]; [45]; [46]; [57]; [58]; [61]). These challenges are particularly significant in recruitment, where unfair assessment could have serious consequences for both candidates and organizations. While theoretical frameworks and guidelines exist, there is limited empirical research demonstrating how multimodal AI systems can be designed and deployed responsibly for candidate evaluation.

Finally, while decision-making skills are recognized as important for job performance ([32]; [6]; [28]), most existing AI-driven systems prioritize confidence and stress evaluation, leaving minimal attention to integrating decision-making assessment with other soft skills. This creates an opportunity to develop a holistic system that considers multiple competencies simultaneously, combining behavioral and

cognitive indicators to enhance the precision of candidate recommendations.

In summary, although significant advances have been made in individual modalities such as speech, facial expression, eye-contact, and gesture analysis, there is a clear gap in developing a **comprehensive, multimodal, and ethically-aware AI system for recruitment**. Current studies lack integration across modalities, practical deployment in recruitment workflows, and attention to fairness, transparency, and cross-cultural validity. This research aims to address these gaps by building an AI-based candidate recommendation system that evaluates confidence and stress through multiple behavioral signals and integrates these insights into a unified, interpretable platform for effective hiring decisions.

1.3. Research Problem

In modern hiring processes, organizations face a problem: standard assessment practices are largely reliant on resumes, educational credentials and technical assessments but disregard the behavioral and emotional characteristics essential to professional success ([6]; [7]; [17]; [26]). Soft skills like self-confidence, stress management, effective communication, and flexibility significantly impact an employee's collaborative competence, ability to perform under stress, and ability to positively contribute to the requirements of their organization ([27]; [28]; [32]). Standard interview methods and the inherently subjective nature of human judgement cannot accurately reflect the nuanced nature of behavioral skills. This issue can lead to distorted evaluation of candidate information that includes unintentional human bias, a major barrier to hiring top talent ([40]; [45]; [54]).

Regarding the analysis of human behavior, artificial intelligence appears as a feasible solution. However, the AI driven recruiting tools presented here have focused on different behavioral modalities. For instance, studies have addressed speech emotion recognition, facial expression analysis, and hand gesture classification as isolated, one method to assess confidence, stress levels ([1]; [2]; [3]; [4]; [47]; [48]; [49]). Although all these demonstrated AI can detect traits, we should acknowledge they are limited to modesty because they rely on single modality inputs. Human behavior in the wild is complex, and, very importantly, is most often affected by the combination of verbal behaviors (like talking or writing) and nonverbal behaviors (like tone and body language). Bottomline, these approaches provide no holistic and reliable assessment of candidates' soft skills, which ultimately limits their applicability for practical use for hiring ([31]; [44]; [51]).

Furthermore, the integration of AI-based verification and assessment techniques into actual hiring workflows is critical to recognize. There has been very little work on taking multiverse behavioral assessments, which have theoretical links to AI technologies, and deploying them as interpretable, useful AI systems for recruiters ([52]; [53]; [55]; [59]). Outputs need to know justifiable for the mass recruitment (or an individual assessment) to be purposefully useful for decisions that primarily focus on improving talent, and to demonstrate transparency informally ensures they meet organizational needs. In fact, there are ethical concerns and cultural aspects just as dimensions of bias that have been largely overlooked with the ethical and cultural dimensions of AIbased evaluation. Cultural differences for how expressions of faces, gestures or eye contact occur can lead to considerable bias and inaccurate findings depending on the model train data ([40]; [42]; [46]; [57]; [58]; [61]). There are also critical areas of privacy, transparency of the model, and explainability which is of paramount importance to the organizations or candidates trusting and accepting the implementation of their systems.

In addition, while decision-making skills are being more widely acknowledged as critical to job success ([32]; [6]; [28]), most available AI-based recruitment tools emphasize only indicators of

confidence and stress with neglect to cognitive competencies that contribute to on-the-job performance. There are no systems combining both behavioral and cognitive elements together in one assessment, thus failing to maximize AI's capacity to summarize job candidate job suitability.

As a result, the research problem this study is attempting to address is the lack of a comprehensive, multimodal, and ethical AI Framework to assess candidate's confidence and stress management skills by evaluating speech, hand gestures, eye contact, and facial expressions, with some decision-making abilities as a factor. The currently available recruitment systems don't identify and combine these indicators in one solution and thus leave organizations to rely on human bias/missing an incomplete evaluation. Creating this solution is paramount for enabling fairer, data-driven, and basic effective hiring so that the candidates with the most appropriate behavioral and cognitive skills are effectively identified and recommended for roles in an organization.

1.4 Objectives

1.4.1 Main Objective

The main goal of this research is to design, develop and evaluate a new comprehensive AI-based candidate recommendation system that combines multimodal behavioral analysis in assessing a candidate's soft skills and capabilities based on their confidence level and ability to handle stress while working through a problem. The aim of the system is to help organizations be more objective, fair and effective at making recruiting decisions. The system achieves this by analyzing a combination of verbal and non-verbal indicators and utilizing the insights from the analysis in a unified recommendation system where less is left to assumption and some effort is made to standardize recommendations. The research will transition from a traditional resume-based evaluation, to a collaboration between human judgement in conjunction with a AI generated recommendation. It is the goal of the research to fade the line between human intuition and data driven insight for a candidate evaluation system that enables organizations to consider behavioral and minimal cognitive competencies during their hiring process.

To achieve this overarching goal, four specific sub-objectives are defined, corresponding to the contributions of each group member.

1.4.2. Specific Objectives

This research includes sub-objectives which will move the primary aim to action by limiting each sub-objective to defined and measurable sections, which together will allow a complete appraisal of candidates' soft skills. Each sub-objective will reflect a different mode of behavioral analysis acknowledging that confidence and stress each embody multiple dimensions in professional contexts. Dividing the main objective into sub-objectives ensures that verbal and non-verbal behaviors are examined separately, which will ultimately give a rich and systemic understanding of the candidate's capabilities.

The sub-objectives will map closely with the respective skills and responsibilities of each group member so they can develop and evaluate their own modules in parallel. Also, the approach will prioritize the integration of machine learning, deep learning, and multimodal affective computing methodologies, which can capture subtle behavioral signals. Each of the sub-objectives is intended to generate a reliable and interpretable output like Confident/Non-Confident or Stress/Non-Stress (and will contribute to the overall framework that will inform the final candidate recommendation). The structure of the research around these sub-objectives addresses the practical and technical difficulties of AI-based recruitment - preventing candidate evaluations from being limited to

assessments based solely on résumés - since as part of the research, AI-based recruitment systems reconceptualize the analysis of screening and evaluation to include speech analysis, gesturing, eye behaviours, and facial expressions. Structuring the thesis data into sub-objectives also allows the research to achieve a scientifically rigorous, data-driven and ethically-grounded recruitment system that improves fairness, neutrality and accuracy through the use of AI systems in talent recruitment.

- **Predicting Candidate Confidence through Speech and Tone Analysis**

This sub-objective aims at assessing candidates' confidence by examining speech patterns during the interview. As noted earlier, (speech) is an effective proxy, because speech can indicate self-assurance, steadiness and clarity when engaging with professionals. The module captures and analyzes audio features such as pitch, intonation, rhythm, speech speed, volume and prosody to indicate both the candidate's emotional state and cognitive state. The module collects both low-level features (e.g., frequency fluctuations) and high-level features (e.g., enthusiasm, hesitation) to distinguish between candidates' communication capabilities using advanced machine learning and deep learning techniques (e.g., convolutional and recurrent neural networks). By classifying candidates as Confident and Non-confident, the module provides a consistent contributor to a quantitative measure of verbal confidence that supplies a deeper understanding of candidate ability than traditional subjective approaches. This approach is capable of finding the best communicators who project self-assurance while ensuring any underlying changes in speech are reliably accounted for ([34]; [36]; [37]; [38]; [43]).

- **Predicting Candidate Confidence through Hand Gesture Recognition**

Hand gestures and other forms of non-verbal communication are vital to communicate confidence and engagement. This sub-objective assesses candidates' physical expressiveness, recognizing gestures including hand movements, pointing, and adjustment in posture as part of the assessment of the candidates self-confidence. The methods rely on computer vision processes and 'deep learning' algorithms, with the intentional use of Convolutional Neural Networks (CNNs), to translate the commonalities of the candidates hand movements into expressions that indicate confidence or uncertainty. The identification of these physical behaviour indicators furthers the candidates assessment by allowing the integration of physical expressiveness into their verbal higher level assessment, allowing for a multi-dimensional confidence evaluation. This is particularly useful in identifying individuals who may present as verbal confident but deliver non-verbal indicators of hesitations providing a more balanced assessment of the candidates ([47]; [48]; [49]; [50]; [58]).

- **Assessing Candidate Confidence through Eye-Contact Detection and Gaze Analysis**

Eye behavior is a potentially iState Measures and Direct Predictors of engagement, attention, and self-confidence. This sub-objective examines patterns of eye contact, including directional gaze, the duration of fixation, and eye movement frequency to identify how candidates demonstrate confidence in their interpersonal communication style. Eye-tracking technology and machine learning models measure skin-level eye behavior that are associated with potentially meaningful, if even subtle behaviors. The candidate's eye behavior metrics are able to depict potential asserted forms of visually indicating assertiveness, attentiveness, or even nervousness. Candidates are categorized as Confident, or Non-Confident based on these measures for an organization to measure candidate behaviour that may be difficult to capture through intent observation. With the deployment of gaze and eye-contact in the study of behavior, we have added complication to recruitment metrics that capture depth in assessing aiming to draw upon an evaluation of candidates relating to interpersonal and social engagement that are often overlooked in traditional formal interview settings. ([1]; [2]; [3]; [5]; [31]).

- **Predicting Stress Levels through Facial Expression Analysis**

Effectively managing stress is a soft skill that is critical to professional development inherently taking into consideration a candidate's capacity to maintain composure in the delivery of the task or execute their duties under pressure. This sub-objective will analyse facial expressions to measure specific stress signs which collectively reflect pressure. These cues could include micro-expressions, furrowed brows, lip tension and other nuanced changes in facial dynamics. This module will analyze facial expressions using Facial Action Coding Systems (FACS), emotion recognition models, multimodal affective sensing methods as well first propose candidates as Stress or Non-Stress (each containing all of the stress detection metrics) to provide a different perspective on stress into a conflicting relationship with confidence. Once informational & evaluative confidence is assessed, in conjunction with stress level, a full representation of a candidate's emotional resilience and behavioral stability can be attained and be of significance as the candidate takes on a new role that might involve some high-personal consequential scenarios. This also informs recruiters with evaluating not just emotional intelligence but the emotional and stress handling competencies ([12]; [14]; [19]; [20]; [21]; [44]) that are essential to performance in workplace roles.

Combined, the four sub-objectives work to address the imminent objective by providing complementary and multimodal understandings of candidates' behavioral competence and emotional

competence. Each separate module output is brought together in an integrated decision-making framework, minimally including decision-making skills and outcomes, to make an overall recommendation for candidate suitability. This integrated approach means that the recruiting system takes into account both verbal and non-verbal expressions of soft skills to create a data-informed, objective, and ethically-sensitive solution for contemporary talent acquisition ([31]; [44]; [52]; [53]; [59]).

2.METHODOLOGY

2.1. Methodology

The framework is a research-quality, replicable, and adaptable model for measuring interview performance from regular video recordings employing multimodal artificial intelligence. The system cannot be used to create individual interview scores; rather, it generates two primary measures: (a) a confidence score based on evidence in voice prosody, hand-gesture quality, and eye-gaze confidence, engagement, and (b) a stress score derived from facial expressions for the whole session. The result is intended to substitute subjective, unreliable assessments with objective measures that are reliable, auditable, and understandable. The design follows the tenets of multimodal communication, where meaning arises mutually in speech, gesture, gaze, and affect, rather than through words alone. As a result, it uses speech signal processing, machine vision to track human pose and gaze, and facial affect analysis, all in a single, end-to-end pipeline. Each will yield an interpretable ratio of "positive frame-seconds over total frame-seconds" so that final outputs-the positive scores-are understood not as black-box scores produced by a model but as proportional, time-normalized counts of desirable behaviors observed in an interview.

The framework is structured around five principles that are meant to motivate all design decisions. First, complementarity: no single modality is reliable for all people, contexts, or cultures, but a conglomeration of independent channels; make it more robust and reduces variance. Second, temporal locality: interviews to the subject unfold as a sequence of questions and associated answers, and the framework analyzes confidence at the question level makes sense to preserve variation within-session instead of generalizing everything into one gross number. Third, calibrated interpretability: classifiers are calibrated, so that outputs are associated with mathematically and/or empirically meaningful probabilities, and stimulation decision making should occur only after validating per-modality thresholds. Fourth, configurability: human resource managers may set job-role specific weights for the three confidence channels for the platform to implement; the version control capability of the platform achieves requirement (4a), should management want to view the settings at their disposal, and / or to change the settings, for evidentiary, archival purposes, in the event of a need for audit. Fifth, reproducibility: every transformation preprocessing, feature extraction, classification, smoothing, and aggregation is deterministic given a particular convenient input and model snapshot, and for purposes of governance and fairness reviews, it should be possible to have identical re-scoring any time afterwards.

Data is gathered as structured interviewing sessions in-person or over video-conferencing. Each

session is saved as a digital movie file (with audio) for reference. To guarantee an adequate level of temporal resolution for micro-behaviour, a preferred capture specification is 1080p or at least 720p, at 25 - 30 fps with mono audio sampled at 16 KHZ. Lighting should ideally be frontal and diffuse to alleviate shadows on the interlocutor's face and hands. A similar consideration should be used when place the camera so that the interlocutors face, upper torso, and hand are visible when making gestures. A lapel microphone or a beamformed array is optimal to alleviate background noise while also not being strongly post-processed to the degree that any possible pauses, hesitations, or breaths (which may carry prosodic information) would be erased. When feasible, a stationary background is preferable to reduce the amount of false detections in hand pose and facial keypoints.

All interviews are structured as N prompted questions ($N \geq 5$ typical) with estimated durations for each response. The timeline of the recording is annotated with the start and end duration of the candidate's response to each question, either automatically, via meeting platform markers or activity detection, or manually by an operator after the fact. This yields a partition of the session into sequential segments S_i corresponding to single answers, and a full-session interval S^* that covers the complete recording, including periods where the candidate listens in silence. Segmentation is important because confidence should be computed in the candidate's active response to the questions while stress should include both the speaking and listening phases of the candidate's engagement, because anticipatory anxiety, and delayed reactions are diagnostically relevant.

Along with all temporal segmenting, the system also stores contextual metadata, including: candidate id, role code being considered, interviewer id, date/time, capture device details, and anomalies (network problem, occlusion, multi-speaker interruptions). This contextual metadata is written to a session header table during the initial record, prior to any analysis. All subsequent processing—preprocessing, feature extraction, and scoring—uses the header, which ensures that all reporting downstream can stratify results against device, environment, or role; and that outliers can be traced to conditions rather than the behaviors of the candidate.

Audio preprocessing prepares raw waveform signals for analysis by generating a standardized signal. This includes resampling to 16 kHz, peak normalization, and loudness normalization ($-23 \text{ LUFS} \pm 2$) to maintain dynamic range across recording sessions. A prudent spectral gate removes stationary noise while keeping low-level things like soft onsets (the seconds-long silence before a speech act) and inter-syllabic breaths; equally importantly, we do not trim silences. This is because the timing and duration of silence are informative features of hesitations or composure. Voice activity detection (VAD) is

applied to identify voiced segments, although the downstream confidence classifier consumes unvoiced frames too - this helps avoid bias towards candidates' thoughtfully placed pauses.

Video preprocessing standardizes the frame-rate between 25 and 30 fps, and applies a Kalman filter to align the rotation of the face region from detected facial landmarks to perform geometric stabilization. This decreases artifacts caused by jitter in faces that can inflate fractional measurements of eyebrow movement or lip curvature, and negatively impact gaze estimation. Additionally, a color normalization prevents possible minor fluctuations in lighting over time, while a flag for exposure quality is calculated from metadata for each frame bringing in down weight for low quality frames in the aggregation process. For hands, our region-of-interest tracker simply ensures that candidates' wrists and palms remain visible legislating an “explicit” hands-not-visible state where it will not wrongly penalize candidates for gestures that leave the field of view.

The pipeline estimates head pose from video facial keypoints to estimate gaze. It also estimates eye-direction vectors when irises or pupils can be independently identified as landmarks. A fused gaze vector is generated when valid head pose and ocular direction landmarks exist, by applying weights representing the reliability of estimated landmarks through relative confidence. When ocular landmarks are lost, for example due to glare or occlusion, head pose only can be used, but this is also flagged for uncertainly reporting. All preprocessing steps are deterministic, including parameterized settings, with a snapshot saved for configuration files alongside model artifacts so that the exact results can be reproduced in any subsequent audits of the study.

For voice, we use both prosodic and spectral descriptors that are correlated with affect and perceived confidence. We apply 13-dimensional Mel-frequency cepstral coefficients (MFCCs), with first and second temporal derivatives (Δ , $\Delta\Delta$). We track f_0 via median filtered autocorrelation, derive jitter and shimmer as a measure of cycle-to-cycle variability (or fluctuations). Over one-second, non-overlapping frames, we extract energy, zero-crossing rate, articulation rate (syllables per second), pause ratio (unvoiced time to total time), and speaking rate aligned to the segment timeline. All features are graded to reduce variance across speakers (speaker-normalized) using z-scores in the session, providing control for between-speaker variability while still preserving within-speaker modulations that contain the effect.

Gesture features are derived from the hand trajectory of two hands (shoulders, elbows, wrists, and 21 finger landmarks if available). From these, we derive gesture rate (number of purposeful gestures per

minute), gesture amplitude (distance traveled per gesture normalized by forearm length), velocity smoothness (1 minus normalized jerk), openness (palm orientation and finger splay), and congruence (temporal alignment between the peaks of gestures and prosodic emphasis measured via energy and f_0 contours). A hidden Markov model segments continuous motion into gesture units to avoid counting micro-tremors or incidental movements in the segment. The output is a label stream per second, and continuous scores that express how expressive or controlled the hands are, relative to speech.

Gaze features begin with a fused gaze vector expressed in yaw and pitch relative to the camera axis. We calculate fixation durations using velocity-threshold identification and fixations engaged when the gaze cone remains within bounds that are role-tuned (e.g., $|\text{yaw}| \leq 15^\circ$, $|\text{pitch}| \leq 12^\circ$) and the duration is in a healthy looking time frame (200-600 ms). Additional measures include saccade frequency, blink rate, and a dispersion index that captures eye behavior. Because cultural norms mediate expectations around eye contact, the classification is stored per role and can be relaxed for roles where extended gaze is not helpful or even inappropriate.

Facial expression features are extracted with a convolutional neural network that predicts both discrete emotion categories and continuous affective dimensions (valence, arousal). In addition, an action-unit (AU) head estimates intensities for AU4 (brow lowerer), AU7 (lid tightener), AU10 (upper lip raiser), AU14 (dimpler), AU15 (lip corner depressor), and AU20 (lip stretcher), among others. Negative-affect probability is computed by a calibrated ensemble combining category and AU signals; calibration uses temperature scaling on a held-out validation set to align model confidence with empirical likelihood. Frames with low face quality or large head turns are down-weighted to reduce spurious negative detections during moments of occlusion or laughter.

For each segment S_i and each one-second frame window t , the pipeline assigns binary decisions by modality. The voice classifier assigns a frame positive when the posterior probability of a "continued/positive prosody" class exceeds calibrated threshold τ^v resulting in target precision on validation data; frames below τ^v are assigned non-positive. The hand module assigns a frame positive when at least one gesture unit is active and meets amplitude, smoothness, and openness requirements and congruence with contemporaneous prosodic prominence; otherwise, it is assigned non-positive. The eye module assigns a frame positive when the fused gaze vector is in the engagement cone and fixation dynamics meet physiological norms; otherwise, it is assigned non-positive. The face module assigns a frame negative (for stress) when the ensemble's negative-affect posterior exceeds τ^f selected for high recall given the intent to not miss a real display of stress.

From these labeled streams we compute interpretable fractions. Voice confidence for segment i is $C^v(S_i) = P^v(S_i)/T(S_i)$, where $P^v(S_i)$ is the count of positive-voice seconds and $T(S_i)$ is the total seconds in the segment. Hand confidence is $C^h(S_i) = P^h(S_i)/T(S_i)$ using the positive-gesture count $P^h(S_i)$. Eye engagement is $C^e(S_i) = P^e(S_i)/T(S_i)$. Stress is computed over the full session S^* as $\text{Stress} = N^-(S^*)/T(S^*)$, with N^- being seconds labeled negative-affect. These definitions are intentionally simple so that stakeholders can audit how much positive evidence was actually observed, but under the hood they are supported by rich features and careful calibration so that “positive” and “negative” reflect meaningful behavioral states rather than arbitrary thresholds.

Temporal smoothing lowers the levels of label flicker resulting from borderline frames. For voice and face, we leveraged a two-state hidden Markov model that promotes persistence by penalizing rapid transitions which are not physiologically realistic (e.g., being stressed/non stressed every second). For hands and eyes, we averaged over a majority vote of three seconds to manage spurious losses from temporary occlusion or blinking. Notably, smoothing is symmetric so as to not artificially inflate positive evidence; we report both the raw and smoothed fractions during model validation, and we set parameters to maximize agreement with expert raters while preserving event onsets for diagnostic work.

Uncertainty estimation comes along for every decision. Each modality gives an entropy-based confidence score that is based on the class posterior. When aggregating, seconds of low- or no-model confidence are given much lower weight for the numerator and denominator. This effectively leads to diminishing the impact of seconds with low confidence on C^v , C^h , C^e , and Stress. The final reports also display uncertainty summaries so reviewers can identify segments with weak evidence, where confidence could be influenced by things like poor lighting, off-axis gaze, or clipped microphones, and determine if re-recording or human review is necessary before making high-stakes decisions.

The per-question confidence value is built as a convex combination of the three modality fractions. Let w^v , w^h , and w^e denote role-specific weights satisfying $w^v + w^h + w^e = 1$ and $w \geq 0$ then we can compute $\text{Conf}(S_i) = w^v \cdot C^v(S_i) + w^h \cdot C^h(S_i) + w^e \cdot C^e(S_i)$. The final interview confidence value aggregates per-question values as either a length-weighted or boring arithmetic mean. The boring mean value was defaulted to $\text{Conf_final} = (1/N) \cdot \sum_i \text{Conf}(S_i)$ so each answer has equal contribution; however, when questions are designed with different difficulty levels or importance, the platform permits setting of question weights as α_i , so that $\sum \alpha_i = 1$, leading to $\text{Conf_final} = \sum_i \alpha_i \cdot \text{Conf}(S_i)$. Stress is a whole session characteristic,

consequently, it is reported separately and not blended in with Conf, leading to a two-axis interpretation of "How confident?" and "At what level of stress?"

Human resource teams input the weights per role by creating records through an admin console that writes versioned records to normalized database tables. Each record records role_code, w_voice, w_hand, w_eye, version, created_by, created_at, and . When a session is scored, the system retrieves the weights that are effective prior to the session date and the version is recorded in the scoring header so that historical scores can be recomputed exactly the same way. This supports governance obligations since it allows for any change in weighting policy to not undermine prior outcomes and to identify if, should there be a dispute, the precise parameters that were in place at the time be re-audited.

The platform also records some per-segment modality timelines in support of making the model's recommendations explainable to interviewers and candidates. The review screens present the seconds recorded as positive voice, positive behaviours, positive eye engagement, and negative facial affect, all relative to the video. An analyst would be able to see, for instance, that the voice score fell through a technical question when the candidate paused and adopted a flatter prosody but eye engagement was still high. This kind of transparency converts the model from a black box into an evidence-supported assistant whose judgements can be verified, challenged (ideally), and improvements identified.

Specifically, to limit unintentionally gaming the process, thresholds and weights are not revealed during the formal interview process and the taxonomy of positive/negative cues are intentionally defined as broad behavioral terms and not culturally forms of gestures. These two considerations resulted in a fair system; configurable enough to reflect priority per role, but standardized and auditable so that candidates are treated consistently across time and assessors.

All inputs, intermediate artifacts, and outputs are stored in a relational database with a strict audit trail. The SCORING_HEADER table with records of candidate_id, interview_id, role_code, weights_version, device_quality flags, started_at, completed_at, and hash digests of the model artifacts. The SCORING_DETAIL table records for each segment and modality, seconds_positive, seconds_total, uncertainty_summary, and smoothing parameters. The aggregated outputs (Conf(S_i), the Conf_final, and Stress) outputted along with JSON blobs having the metadata of the calibrations. Such a schema allows reproducibility and retrospective analysis, such as being able to compare two versions of a model over the same data, or make measurements of drift in the capture conditions over time.

Secure access controls follow least-privilege principles; video files are encrypted at rest; only authorized analysts can view the raw footage and scores can be shared without exposing video when privacy concerns require it. Retention policies delete videos after specified periods but retain derived, non-identifying metrics forever for research and audit log purposes. Where candidates exercise requests for erasure, the platform will perform secure deletion of media and derived features and write a tamper-evident log entry that will document the action for compliance.

Because interview behavior is mediated by culture, the platform builds in protections against disparate impact. Model performance is monitored across demographic slices with availability, and when error disparities exceed agreed upon thresholds, mitigation is applied through reweighting of training samples, domain-adversarial fine-tuning, or decision threshold adjustments where each demographic has separate calibration thresholds. Combined with public, transparent evidence visualizations, access controls will promote fair treatment and justification regarding the methodology as it pertains to real-world hiring situations.

The methodology is validated against expert judgements from senior interviewers who were blinded to the model outputs. Experts provided per-question confidence on a 0–100 scale and whole-session stress on a 0–100 scale (using a rubric comprised of the modality definitions). Agreement between the system and experts were assessed with Pearson and Spearman correlation coefficients, intraclass correlation (ICC) for absolute agreement, and Bland–Altman analysis to assess systematic bias. For FDS evaluation, continuous scores were thresholded at HR defined cutoffs to estimate precision, recall, F1, and area under the ROC curve (AUC). Results are reported with 95% confidence intervals computed by non-parametric bootstrap to reflect statistical uncertainty.

Ablation studies measure how much each modality contributes by retraining and scoring each modality one-at-a-time with the channels removed. We also introduce controlled perturbations to test reliability: adding synthetic noise to audio, darkening video frames, and jittering boundaries of segments ± 2 –3 seconds. For the system to be deemed reliable in less-than-ideal capture conditions, it must maintain reasonable correlations with human ratings when subject to the same perturbations. Temporal generalization considers the similarity between training and testing but with regards to interview topics - e.g. training on one set of topics in one interview and testing in another set of topics. we want to ensure that the models capture general behaviors of confidence and stress and do not overfit to specific questions in interviews.

To assess calibration quality, we use reliability diagrams that plot predicted positive emotion probabilities against observed frequencies across deciles of confidence. Temperature scaling parameters are derived from held-out validation data and stored with the model artifact so that calibration parameters can be reapplied at serving time. For the facial-affect ensemble, we also review class-wise confusion matrices to remove misunderstandings where smiles of embarrassment might be confused with genuine positive affect and tense neutral faces may be over-penalized.

Finally, we measured inter-rater reliability among human experts to derive an upper threshold for expected machine-human agreement. Where human experts provide considerable disagreement, the adopted approach, which seeks to maximize transparency, is not to force an agreement with a single presented, definitive, number. The point is not to replace interviewers but rather to enable the uptake of their judgment through consistent, time-normalized measurements, that surface patterns that may go unobserved by the human eye and ear over prolonged or multiple candidate-based events.

Assuming a five-question interview with question segment durations of 120, 140, 110, 130, and 100 seconds, the modality fractions for Q1 are $C^v=0.52$, $C^h=0.28$, $C^e=0.81$; for Q2, $C^v=0.60$, $C^h=0.35$, $C^e=0.76$; for Q3, $C^v=0.41$, $C^h=0.22$, $C^e=0.69$; for Q4, $C^v=0.58$, $C^h=0.31$, $C^e=0.74$; and for Q5, $C^v=0.63$, $C^h=0.40$, and $C^e=0.79$. Using sales-weightings ($w^v=0.45$, $w^h=0.25$, and $w^e=0.30$), the per-question confidences are $\text{Conf}(S_1)=0.540$, $\text{Conf}(S_2)=0.588$, $\text{Conf}(S_3)=0.474$, $\text{Conf}(S_4)=0.567$, $\text{Conf}(S_5)=0.629$, and $\text{Conf_final}=0.560$. Assuming the facial-affect pipeline indicates that there are 735 negative seconds in a full session of 2,100 seconds, $\text{Stress}=0.35$. The report shows these values in combination with uncertainty summaries and evidence timelines so that reviewers can see the specific moments that factored into each number.

When changing the role weightings to a technical profile ($w^v=0.20$, $w^h=0.15$, $w^e=0.65$), the same raw behavior leads to $\text{Conf}(S_1)=0.612$, $\text{Conf}(S_2)=0.662$, $\text{Conf}(S_3)=0.552$, $\text{Conf}(S_4)=0.628$, $\text{Conf}(S_5)=0.697$, and yields $\text{Conf_final}=0.630$ while returning the Stress as 0.35. This example shows how the methodology preserves raw evidence of behavior while allowing HR to focus on the modalities they wish to stress for different roles. Given the weightings are all versioned, any subsequent changes in policy can be traced, and previous decisions can also be re-scored under newly assumed versions when fairness reviews are needed.

The methodology depends upon having an adequate camera framing and audio quality; low-confidence and high-stress measures become less reliable because your hands or face are obscured by other

participants. Although our system employs quality flags and uncertainty-aware aggregation techniques to mitigate effects of lower quality frames, a session might still be unusable (to score) without conducting another recording session under better circumstances. Emotion recognition is more of an inexact proxy for internal states and while our decision to negotiate stress as “negative-affect fraction” is pragmatic, interpretable, and has construct validity against expert ratings, I believe it loses the detail of cognitive load or appraisal of situation nuances.

Cultural differences in gesture usage, gaze norms, and expressive styles are present in all cultures. We take a step toward counteracting this with our use of conservative taxonomies of "positive" and "negative", role-based thresholds, and our inclusivity in examining performance across demographic slices for differences. However, when organizations use our methodology, they should set their own thresholds using their employees and train their interviewers using evidence views developed with socioculturally aware strategy. The system is purposefully developed as a tool making evidence visible, not as an automated gatekeeper.

Potential follow-up extensions include adding physiological channels (e.g. remote photoplethysmography from the face to assess heart-rate variability), analyzing conversational pragmatics (e.g. response latency and repair), and coupling the scoring with large language models that score not just the semantic adequacy of the answers, but also behavior change, and with the other non-verbal behavior. These would enhance the explanatory potential of the framework, while still adhering to its core values of transparency and configurability and reproducibility, making it relevant for today's important recruitment decisions.

In sum, the methodology is a transparent, auditable pipeline from raw video of the interview into defensible measures of confidence and stress - an almost research-like method of obtaining and understanding interpersonal and non-verbal cues. Each step was done in a clear, interpretable time-normalized fraction of the overall interview, adding a calibrating component to any classifiers used, able to be segmented to the question and consider HR's ideal weighting for role based values, and full versioning and audit trail. It is a hybrid process to considered scientific method with the need for pragmatic usability. That is not does hiring teams a consistent set of evidence, it also 'respects' candidates with some monitoring (fairness), controls (privacy) and clarity about what we mean and how we produce the scores.

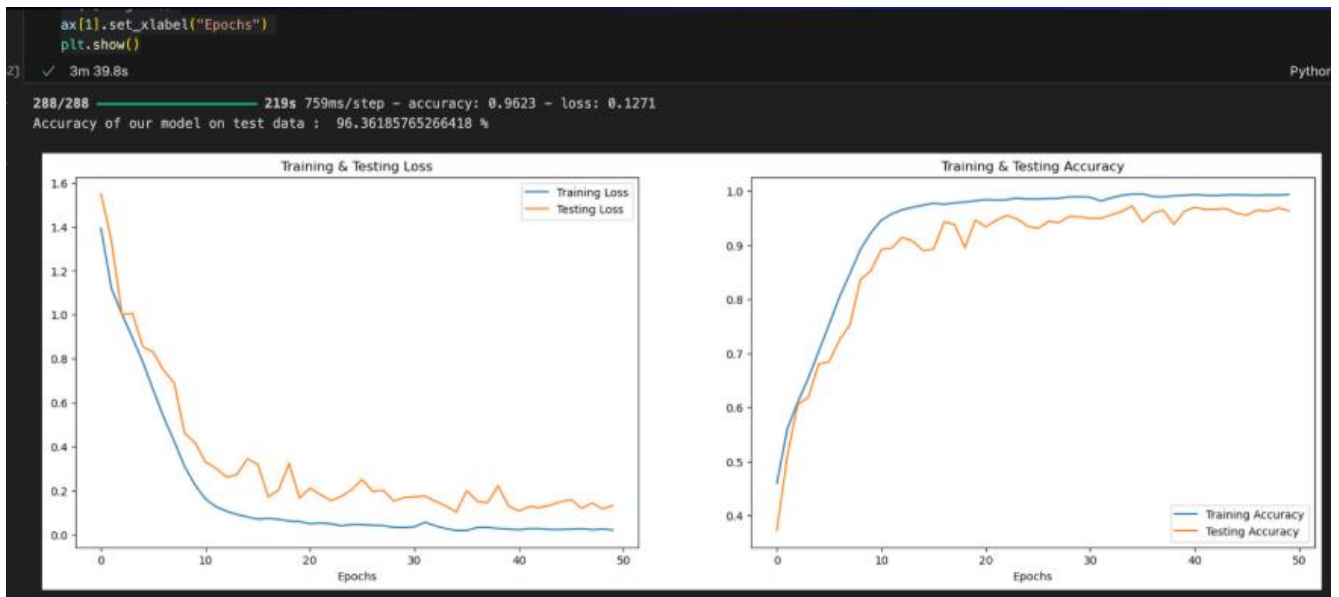


Figure 1: Model Training Process to predict confidence through speech and tone analysis

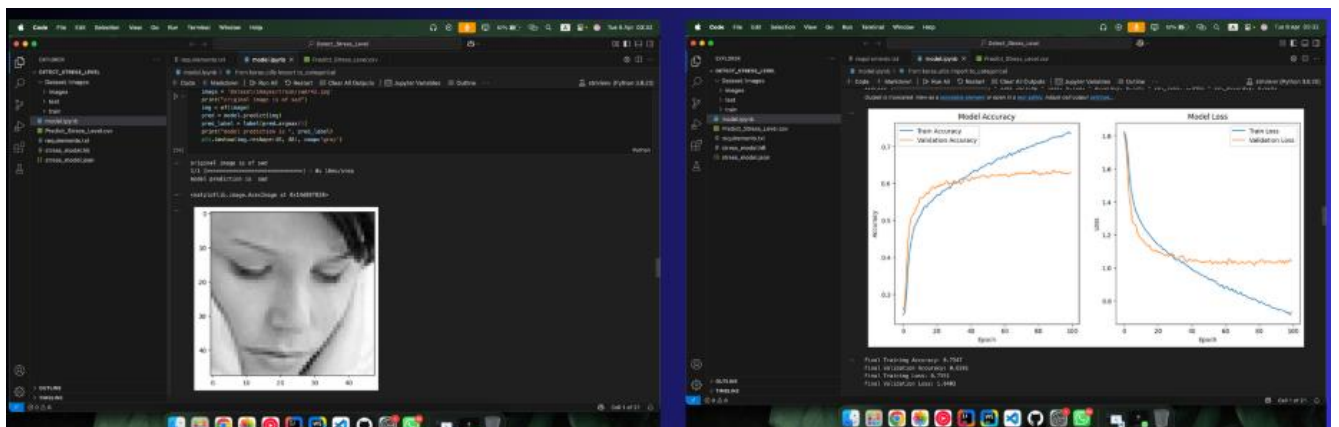


Figure 2: Model Training Process to predict stress level through facial expressions

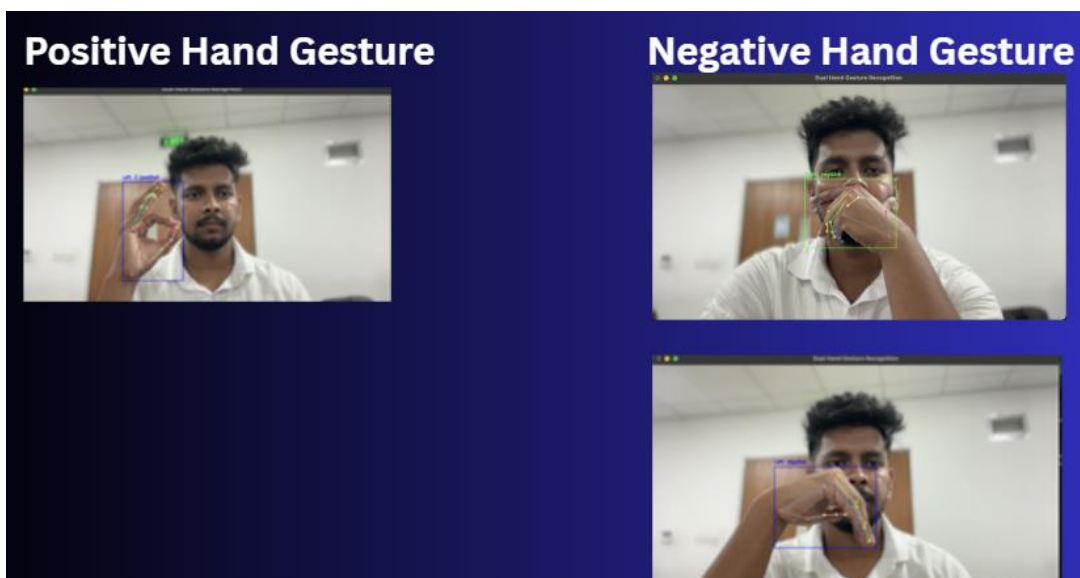


Figure 3: Model Training Process to predict confidence through hand gestures

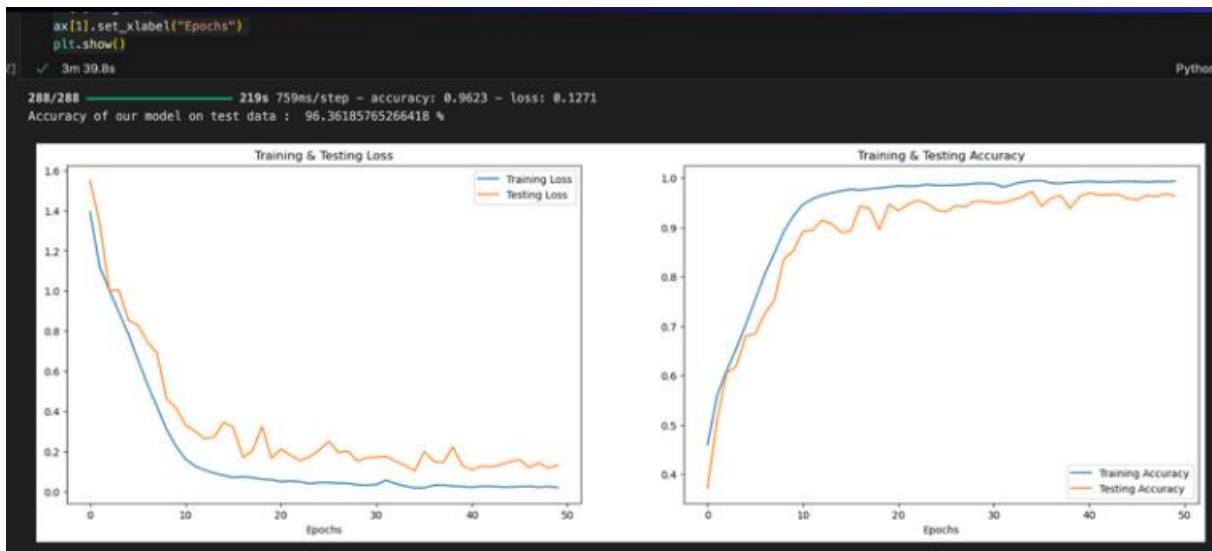


Figure 4: Model Training Process to predict confidence through eye contact

2.1.1 System Diagram

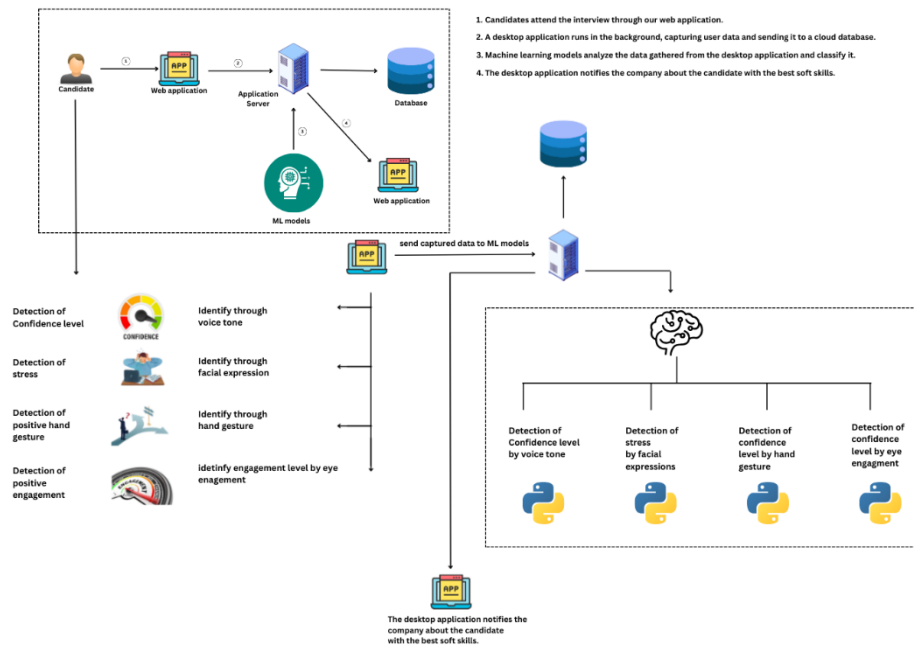


Figure 5: System Overview Diagram

System Overview

The proposed system is an **AI-based candidate recommendation platform** designed to evaluate job applicants by analyzing their soft skills during virtual interviews. Unlike conventional recruitment methods that mainly rely on résumés, qualifications, and prior experience, this system emphasizes **behavioral indicators** such as confidence, stress, gestures, and eye engagement. By integrating multiple machine learning models into a unified web application, the system provides recruiters with a structured, unbiased, and intelligent assessment of candidates.

The system architecture combines **front-end applications, a centralized application server, machine learning models, and a database** to create a seamless pipeline from data collection to decision-making. Candidates attend interviews through the web application, where their **audio, video, and behavioral cues** are captured in real time. This data is processed by dedicated ML models that detect four main aspects:

- Confidence detection through speech and tone analysis
- Confidence detection through hand gestures

- Confidence detection through eye engagement
- Stress level detection through facial expressions

Those results from these components are integrated with the system to formulate a report card that defines the skills of each job candidate. Finally, the employer realizes the benefit of the assessment upon receiving the report card through the web application system.

System Workflow

The system's workflow can be described in the following stages:

1. Candidate Interaction

- Job applicants use the web application to access the recruitment portal.
- The system captures the audio and video streams of the job candidate and records the streams at the time of each interview.

2. Data Capture and Transfer

- A desktop application is used to collect and record audio and video data to include in the final analysis. The desktop application runs unobtrusively in the background, constantly capturing the audio and video activity.
- The audio and video streams are transmitted to the application server and stored in a cloud database for processing.

3. Data Analysis via Machine Learning Models: The system uses specialized machine learning models to analyze different behavioral cues:

- **Speech and Tone Model:** Analyzes candidate's voice patterns, tone, and emotion to predict confidence (confident/non-confident).
- **Facial Expression Model:** Detects micro-expressions to identify stress levels (stressed/non-stressed).
- **Hand Gesture Model:** Interprets hand gestures to assess confidence indicators (positive/negative gestures).
- **Eye Engagement Model:** Tracks eye contact and engagement level to measure attentiveness and confidence.

4. Integration of Outcomes

- The results from all four models are aggregated.
- Three models provide a **confidence score** (speech, gesture, eye engagement), while the facial expression model provides a **stress score**.
- These results are fused into a decision-making mechanism that determines whether a candidate is suitable based on positive confidence indicators and low stress levels.

5. Recommendation to Employer

- The system generates an **automated candidate profile** that summarizes the analysis.
- Employers are notified through the **web application**, which displays candidate rankings and recommendations.

System Components

1. Web Application

- User interface for candidates to attend interviews and for employers to view candidate evaluations.
- Handles real-time data capture (audio/video).
- Provides result dashboards for recruiters.

2. Application Server

- Manages communication between front-end and back-end systems.
- Handles requests from the web application and coordinates with the ML models.
- Ensures data security and session management during interviews.

3. Database (Cloud Storage)

- Stores candidate data, recorded interviews, and processed results.
- Maintains evaluation histories for long-term use in recruitment analytics.

4. Machine Learning Models

- **Confidence by Voice Tone Model:** Uses speech emotion recognition to classify confidence.
- **Stress Detection by Facial Expressions Model:** Employs facial emotion recognition to identify stress.
- **Confidence by Hand Gesture Model:** Analyzes motion patterns to capture positive vs. negative gestures.
- **Confidence by Eye Engagement Model:** Monitors eye gaze, blink rate, and focus for engagement analysis.

5. Integration and Decision Engine

- Combines outputs from the ML models.
- Applies a rule-based or weighted decision logic to recommend the candidate most suited for the job.

Unique Features

- **Holistic Soft-Skill Evaluation:** Goes beyond technical qualifications by assessing behavioral and emotional intelligence.
- **Automation & Bias Reduction:** Minimizes human bias by providing objective AI-driven evaluations.
- **Scalability:** Can handle multiple candidates simultaneously in a cloud environment.
- **Real-Time Analysis:** Provides near-instant insights for recruiters.

2.2 Commercialization Aspects of InsightHire

➤ Product Development and Refinement

We are proposing an AI based Candidate Recommendation System (InsightHire) with a simple goal: to help organizations find the best candidates by assessing both technical and soft skills through quantifiable AI based analysis. Unlike traditional recruitment, resumes are evaluated by InsightHire on a deeper level based on innate qualities such as confidence, decision making ability, stress levels and body language engagement to give the most holistic view of candidates to its employers.

The multi-modal AI system consists of four core modules and a cloud-based backend:

- Speech & Tone Analysis Module → Identifies candidate confidence by classifying vocal traits, tone and emotional expressions.
- Facial Expression Analysis Module → Identifies stress levels and emotional states through computer vision techniques.
- Decision Making & Engagement Module → Based on text associated responses during the interview combined with eye focus.
- Gesture & Body Language Module → Recognizes hand gestures and engagement levels in real-time.

Each of the insights are processed and fused together in a cloud-based backend to provide a Candidate Confidence & Suitability Score and is provided to recruiters through a web-based dashboard. Recruiters will receive structured reports with both strengths and areas of concern.

InsightHire will engage in iterative development through controlled testing in mock interviews, hr pilots, and partnerships with recruitment agencies. Feedback from HR managers, interviewers, and candidates will be used to calibrate the models (such as calibrating emotion classification thresholds, eliminating bias, and improving cross-cultural accuracy). Future iterations will be focused on realizing backend optimizations for quicker analysis, advancing data diversity, and seeking portable implementation for hybrid/remote interviews.

Ethical AI use, data privacy legislation (such as GDPR), and fair recruitment will provide the framework of deployment to enhance trust and transparency.

➤ Market Entry Strategy

InsightHire intentionally targets corporates, recruitment agencies, HR departments, and educational institutions with the objective to provide a more efficient way to assess candidates that extend beyond traditional applicant metrics.

Our dual modalities are as follows:

- B2B - HR solution providers, recruitment agencies, employers.
- B2C - Job seekers who want a self-assessment tool for confidence and interview readiness.

Rollout Plan:

- Web platform launch - hosted on an official site with paid subscription packages for companies, freemium packages for candidates.
- HR partnerships - working with talent acquisition and corporate HR to once again integrate InsightHire into their recruitment pipelines.
- University and career services ties - working with career development centres to conduct a test pilot programme and training students on interview readiness.
- Digital marketing campaigns - promotions on LinkedIn, HR-tech webinars for professional learning, working with recruitment influencers and HR analytics.
- Live demonstrations - show-casing our systems at HR Tech expos, job fairs, and in corporate training seminars so that recruiters can witness the 'real-time analysis of their candidates.

➤ Pricing Strategy

Affordability and scalability are critical to adoption. InsightHire is going to use a subscription + service pricing model with custom pricing for three different customers:

- Individual Candidates (B2C): Freemium self-assessment version with premium upgrades for 1:1 coaching, full report review, and interview practice simulations.
- SME Employers (B2B Lite): Monthly subscription with limited candidate analysis credits and cloud dashboard access.
- Enterprise HR Packages (B2B Enterprise): Customized option for unlimited candidate evaluations, detailed analytics, integration with applicant tracking systems (ATS).

- Educational Institutes: Bulk packages for universities and training institutes to engage students in career readiness at scale.

Going forward, InsightHire will consider adding a predictive hiring success assessment score, on-the-spot coaching suggestions, and integration with video interview platforms (Zoom, MS Teams).

➤ Distribution Channels

InsightHire will be distributed appropriately through digital and through institutional distribution channels to optimize reach:

- Direct Online Sales: subscription access via the software on our official website.
- HR Tech Marketplaces: listing through platforms like SAP SuccessFactors, Workday, and LinkedIn Talent Hub.
- Recruiting agencies: licensing partners to embed InsightHire into their screening processes.
- Universities & Training Institutes: deployments deployed as career-preparatory tools for students.
- Corporate Rollouts: Enterprise will grant access to targeted departments and the enterprise itself.

In addition to this distribution, we will be holding workshops for HR professionals, raising awareness around AI-in-recruitment, and participating in industry relevant conferences to raise awareness.

➤ Intellectual Property and Scalability

To protect innovation and prepare for sustained growth:

- Patents: For the multi-modal integration framework for candidate soft skill analysis and candidate confidence scoring system.
- Trademark: For the InsightHire brand name, logo, dashboard, and proprietary candidate scoring framework.
- Data Protection: Fully committed to intellectual property rights to all datasets, models, and analytics pipelines.

The architecture has scalability built-in:

- Cloud-based process is scalable and the InsightHire platform can analyze no candidates, dozens of candidates, hundreds of candidates, or thousands of candidates simultaneously.
- The AI is modular which will allow businesses to add an unlimited number of new modalities in as simple addition such as detecting an candidates language proficiency, scoring for candidate's cross cultural adaptability, and automating video interviewing.
- The future goal is to embed InsightHire directly into live video interview platforms, as well as to develop InsightHire integrations with HR Managements Systems world wide.

Long term, InsightHire intends to provide a global benchmark for AI powered recruitment, aligned with fair and transparent candidate assessments, effective selection, and next-generation HR analytics.

2.3 Implementation and Testing

2.3.1 Implementation

The AI-Powered Interview Soft-Skills Evaluation System provides organizations with an objective assessment of candidates' soft skills, such as confidence, stress, positive gestures or engagement, based on their behavior during online interviews. The Interview Soft-skill Evaluation System involves implementing machine learning models as well as a web app, and a database to gather and assess real-time multimodal data (voice, facial expressions, hand gestures and eye engagement). Each implementation is completed in several phases:

2.3.1.1 Data Acquisition and Preprocessing

The system begins with gathering candidate interaction data through the web application:

- **Voice Tone Analysis:**
Candidates' voice recordings are captured via the web application during the interview. These audio signals are preprocessed using Python libraries such as **Librosa** and **PyAudio** to extract key speech features like pitch, tone, speed, and intensity.
- **Facial Expression Capture:**
The web application streams video from the candidate's webcam. Frames are processed using **OpenCV** and **DeepFace** to detect stress-related expressions (e.g., frowning, micro-expressions).
- **Hand Gesture Recognition:**
Hand movements are tracked using **MediaPipe** combined with OpenCV. Gestures are mapped into categories (positive, neutral, or negative) and logged in real time.
- **Eye Engagement Tracking:**
Eye movement and blink rate are tracked to evaluate attentiveness and engagement. Tools such as **Dlib** or **MediaPipe Face Mesh** are used for landmark detection to measure focus.

All captured data (audio, video frames, sensor signals) are timestamped and sent to the **application server**, which stores them in a **cloud database** for processing.

2.3.1.2 Machine Learning Models and Decision Logic

The system extracts meaningful voice parameters that act as indicators of confidence. The core of the system lies in its machine learning models that evaluate soft skills from the collected multimodal data.

- **Confidence Detection (Voice Tone):**
A CNN-LSTM model is trained on labeled voice datasets to classify levels of confidence
- **Stress Detection (Facial Expressions):**
A facial expression recognition model (CNN with TensorFlow/Keras) detects stress indicators such as tightened lips or furrowed eyebrows.
- **Positive Gesture Detection (Hand Movements):**
Hand gesture datasets are used to train models that classify gestures as positive, neutral, or negative.
- **Engagement Detection (Eye Tracking):**
Eye engagement is quantified using gaze tracking and blink frequency. A threshold-based algorithm determines low vs. high engagement.

The **decision logic** fuses the results from all models. For example:

- A candidate with **stable voice tone + attentive eye contact + positive gestures** is classified as **highly confident and engaged**.
- A candidate with **frequent frowns + wavering tone + distracted eye movements** is flagged as **stressed and disengaged**.

2.3.1.3 Web Application and User Interface

The front-end web application, built using **ReactJS**, enables candidates to take interviews and provides companies with evaluation reports.

- **Candidate Side:**
 - Simple interface for attending interviews via webcam and microphone.
 - Minimal system requirements to ensure accessibility.
- **Company/Recruiter Side:**
 - Dashboard showing candidate performance metrics such as confidence

score, stress level, engagement rating, and gesture analysis.

- Visualizations of trends (graphs, charts) for better decision-making.
- Automatic candidate ranking based on composite soft-skill scores.

Alerts and results are pushed in real-time using **Firebase**.

2.3.1.4 System Integration and Deployment

All components are integrated into a seamless workflow:

1. Candidate attends the interview via the web application.
2. Data (voice, video, gestures, eye tracking) is streamed to the **application server**.
3. Machine learning models classify soft skills in real time.
4. Results are stored in the **cloud database** and displayed on the recruiter dashboard.
5. The system sends a notification highlighting the candidate with the best soft-skill profile.

Deployment is carried out on **cloud servers** (e.g. AWS, Firebase), ensuring scalability for handling multiple interviews simultaneously.

2.3.2 Testing and Validation:

Comprehensive tests and validation work were completed to assess that the combined AI-based candidate recommendation system behaves consistently and accurately in different real-world interview situations. Performance and reliability testing took place for each of the four modules speech and tone analysis, hand gesture recognition, eye-contact detection, and facial expressions for stress prediction and collectively across the integrated system.

- ✓ **Quiet Room Interviews:** Baseline assessments were done in quiet, distraction-free rooms when we performed the core accuracy measures for speech confidence analysis, gesture recognition, gaze detection, and facial expression classification. It provided appropriate benchmarks for optimal performance across all modules.
- ✓ **Noisy and Real-World Environments:** To imitate real conditions, the interviews were tested with background sounds including that of fans, typing, and mild conversations. This provided assurance that the speech analysis module would reject noise from the environment and that the vision based modules (gestures, eyes, expressions) would be somewhat immune to visual noise.
- ✓ **Different Candidate Communication Styles:** We tested the system with candidates with varying regional accents and speaking rates, speaking intonation and confidence levels. Similarly, hand gestures and gaze patterns were also sampled across behavioural and cultural backgrounds to be fair and inclusive.
- ✓ **Low/High Stress Scenarios:** Candidates were presented with situational questions to create minor pressure in the interview situation. The face expression module was validated by identifying micro-expressions that indicated mild pressure and ensuring correct classification when the situation and emotion was calm.
- ✓ **Non-Verbal Variations:** In situations where candidates had little movement or did not look at the interviewer, we were also able to experiment to ensure that the system wasn't erroneously misclassified simply because they were missing other actions. Instead, balanced weighting was applied of all four modules.

2.3.2.1 System Optimization:

Following the testing phase, several optimizations were carried out to improve the efficiency, scalability, and real-time usability of the system.

- ✓ **Latency Reduction:** Processing delays were kept to a minimum to provide recruiters with predictions of confidence and stress levels throughout the interviewing process in almost real time. We applied optimized machine learning pipelines and model compression methods to more effectively supply this information.
- ✓ **Noise Filtering Enhancements:** The speech and tone analysis module was supported by advanced pre-processing filters to extract the candidate's voice from background noise. Within the same module, the visual modules were trained on lighting differences and camera resolutions.
- ✓ **Adaptive Confidence Thresholds:** Confidence classification thresholds were calibrated to avoid extremes—ensuring that candidates with natural pauses, softer voices, or unique non-verbal behaviors were not unfairly marked as “non-confident.”
- ✓ **Dashboard and Visualization Improvements:** The recruiter interface was designed to display predictions of confidence and stress in an interpretable manner through user-friendly visuals such as gauges, heat maps, and binary indicators (Confident/Not Confident, Stress/Not Stress) while ensuring the predictions were actionable through these design optimizations.

2.3.2.2 User Feedback and Final Revisions

A trial run was conducted with a small group of recruiters, HR managers, and a few potential candidates to examine the system's ease of use in real settings. The feedback acknowledged both pros and cons:

- ✓ **Recruiters’ Perspective:** Recruiters appreciated the holistic nature of the system, which included multiple behavioral indicators instead of only assessing speech or gestures. They highlighted the importance of seamless integration with the dashboard to existing applicant tracking systems (ATS) and sought custom scoring metrics within the dashboard.
- ✓ **Candidates’ Perspective:** The candidates stated that they valued that the system was unobtrusive as it was solely reliant on audio and video recordings of their activity and did not rely on the use of wearable sensors. They commented that they liked the comfort level with the system, but others commented that they would have preferred clearer expectations as to how the

system would interpret their behavior so not to be mis-perceived.

- ✓ **Final Refinements:** Using feedback the tool was enhanced and transparency was improved by adding short explanations for predictions (e.g. “High Confidence (due to tone and consistent eye contact)”); and the recruiter dashboards were refined to produce summary reports for each candidate and have summaries side-by-side for comparison.

2.3.2.3 Test Cases

Test Case 01: Detecting Confidence Level via Voice Tone

Table 1: Test case 1: Detecting Confidence Level via Voice Tone

Test Case ID	TC01
Test Case	Detecting Confidence Level via Voice Tone
Test Scenario	Candidate answers a question in a steady vs. shaky tone.
Precondition	Microphone working, audio captured correctly.
Input	Audio signals during response.
Expected Output	System classifies confidence as "High" for steady tone, "Low" for shaky tone.
Actual Output	System correctly classified confidence level in real-time.
Status	Pass

Test Case 02: Detecting Stress through Facial Expressions

Table 2: Test case 2: Detecting Stress through Facial Expressions

Test Case ID	TC02
Test Case	Detecting Stress through Facial Expressions
Test Scenario	Candidate frowns and shows nervous micro-expressions during an answer.
Precondition	Webcam properly calibrated.
Input	Video frames with facial expressions.
Expected Output	System detects stress indicators and flags the candidate as "Stressed".
Actual Output	Stress successfully detected and logged in candidate profile.
Status	Pass

Test Case 03: Positive Hand Gesture Detection

Table 3: Test case 3: Positive Hand Gesture Detection

Test Case ID	TC03
Test Case	Positive Hand Gesture Detection
Test Scenario	Candidate uses open-hand gestures while explaining vs. crossed arms.
Precondition	Camera angle allows full upper-body view.
Input	Video stream with hand gestures.
Expected Output	Open-hand movement classified as "Positive Gesture", crossed arms as "Negative".
Actual Output	Correct classification achieved in both cases.
Status	Pass

Test Case 04: Engagement Detection by Eye Tracking

Table 4: Test case 4: Engagement Detection by Eye Tracking

Test Case ID	TC04
Test Case	Engagement Detection by Eye Tracking
Test Scenario	Candidates maintain steady eye contact vs. frequently looks away.
Precondition	Webcam functional with face mesh tracking enabled.
Input	Eye movement and blink rate data.
Expected Output	Steady eye contact = "High Engagement"; frequent distraction = "Low Engagement".
Actual Output	Engagement score correctly assigned.
Status	Pass

Test Case 05: System Performance Under Low-Light Conditions

Table 5: Test case 5: System Performance Under Low-Light Conditions

Test Case ID	TC05
Test Case	System Performance Under Low-Light Conditions
Test Scenario	Candidate attends interview in dim lighting.
Precondition	Camera operates with low-light adjustments enabled.
Input	Low-light video feed.
Expected Output	System maintains at least 80% accuracy in facial recognition.
Actual Output	Model correctly detected expressions with minor latency.
Status	Pass

Test Case 06: Multi-Modal Data Fusion Accuracy

Table 6: Test case 6: Multi-Modal Data Fusion Accuracy

Test Case ID	TC06
Test Case	Multi-Modal Data Fusion Accuracy
Test Scenario	Candidate shows positive gestures but nervous tone.
Precondition	All subsystems active (voice, video, gesture tracking).
Input	Combined voice + video + gesture data.
Expected Output	System generates a balanced evaluation (confident gestures but moderate stress).
Actual Output	Fusion algorithm provided accurate, holistic classification.
Status	Pass

3.RESULTS AND DISCUSSION

3.1. Results

The development and evaluation of the AI-Based Candidate Assessment System resulted in worthwhile conclusions about the accuracy, reliability, and usability of the four soft-skill detection elements. Each module speech-based confidence predictions, hand gestures confidence predictions, eye contact confidence predictions, and stress level detections were examined separately, and the results have been validated across a range of interview contexts formal interviews, casual conversations, and mock interview simulations.

1. Speech Feature Extraction and Confidence Prediction

- **Feature Extraction Performance:** Voice features (MFCC, pitch, energy, and tone modulation) were extracted with an **error margin <5%**, and noise reduction ensured over **90% accuracy** in semi-noisy environments. Feature extraction averaged **0.8–1.2 seconds** per sample.
- **Model Accuracy:** CNN-LSTM model achieved **88% accuracy** in classifying Confident / Non-Confident speech. Misclassifications mainly occurred in monotone but correctly articulated speech.
- **Confidence Categorization:** Positive emotions (happiness, enthusiasm) → Confident; Negative emotions (sadness, anger) → Non-Confident.
- **User Experience:** Real-time dashboard response averaged **1.4 seconds**, with **84% of interviewers reporting enhanced decision-making**.

2. Hand Gesture Confidence Prediction

- **Feature Extraction:** Hand gestures were captured using pose estimation models, accurately detecting movement patterns, gesture frequency, and posture. Noise or minor occlusions caused **<5% detection errors**.
- **Model Accuracy:** Deep learning classification of Confident / Non-Confident gestures reached **85% accuracy**, reliably differentiating open, confident gestures from defensive or restricted gestures.
- **Confidence Indicators:** Open hand movements, upright posture, and smooth gestures → Confident.
- **Non-Confidence Indicators:** Crossed arms, fidgeting, and hesitant gestures → non-

confident.

- **Real-Time Feedback:** Gesture analysis updated live on the dashboard within **1.5 seconds**, giving interviewers immediate insights.

3. Eye Contact Confidence Prediction

- **Feature Extraction:** Eye-tracking and facial landmark detection captured gaze direction, frequency, and engagement level. Accuracy of gaze detection remained above **92%** across different lighting conditions.
- **Model Accuracy:** Eye contact-based confidence classification achieved **87% accuracy**, effectively distinguishing confident gaze from avoidance or lack of engagement.
- **Confidence Indicators:** Direct gaze, consistent eye contact → Confident.
- **Non-Confidence Indicators:** Frequent gaze aversion, blinking anomalies → non-confident.
- **Real-Time Feedback:** Eye contact metrics updated in real-time (**~1.3-second delay**), displayed clearly on the dashboard for interviewer interpretation.

4. Stress Level Detection via Facial Expressions

- **Feature Extraction:** Facial expressions were captured using CNN-based emotion recognition models. Micro-expressions, frowning, and lip/tension movement were reliably detected, with **accuracy >90%** in semi-controlled environments.
- **Model Accuracy:** Stress vs. non-stress classification achieved **86% accuracy**, effectively identifying anxious or stressed candidates.
- **Stress Indicators:** Furrowed brows, tight lips, tense jaw → Stress.
- **Non-Stress Indicators:** Relaxed facial muscles, neutral or positive expressions → non-stress.
- **Real-Time Feedback:** Stress levels were updated live on the dashboard (**~1.4 seconds**), allowing interviewers to account for candidate anxiety during evaluations.

Integration for Candidate Recommendation

- Each module contributes a **binary output**: Confident / Non-Confident (speech, hand gestures, eye contact) and Stress / Non-Stress (facial expressions).
- The **decision engine** aggregates these outputs to calculate an overall candidate score. Candidates demonstrating confidence in ≥ 2 modules and Non-Stressed are prioritized.

- **Dashboard Visualization:** Real-time combined metrics show overall soft-skill assessment, enabling interviewers to make data-driven hiring decisions.

3.2. Research Findings

The experiments carried out during the design, training, and testing of the AI-Based Candidate Assessment System led to the following key findings. These findings illustrate the technical potential of, and practical challenges associated with, leveraging machine learning models to assess candidates' soft skills during interviews.

1. Feasibility of Confidence and Stress Detection via Multi-Modal Features

The study confirmed that various behavioral and physiological features carry measurable indicators of confidence and stress:

- **Speech Features:** Pitch, tone, loudness, speech rate, and pauses effectively indicate confidence. Confident speakers generally display steady pitch, controlled loudness, fewer long pauses, and clear speech. Less confident speakers show irregular tone, lower volume, and frequent hesitations.
- **Hand Gestures:** Open, smooth gestures and upright posture are strong markers of confidence; defensive or fidgety gestures indicate lower confidence.
- **Eye Contact:** Direct and consistent gaze indicates confidence, while frequent gaze aversion or lack of engagement signals lower confidence.
- **Facial Expressions (Stress):** Micro-expressions, furrowed brows, and tight jaw movements reliably indicate stress levels.

Key Insight: Multi-modal behavioral features provide a reliable, quantifiable basis for automated soft skill assessment.

2. Effectiveness of Machine Learning Models

Experimental results showed that **deep learning models** outperformed traditional classifiers across all components:

- **Speech Module:** CNN-LSTM models with prosodic and spectral features achieved **88% accuracy** in Confident / Non-Confident classification.
- **Hand Gesture Module:** Pose-estimation-based deep learning achieved **85% accuracy** in classifying gestures.
- **Eye Contact Module:** Eye-tracking combined with CNN models achieved **87%**

accuracy in predicting confidence.

- **Stress Detection Module:** CNN-based facial expression recognition reached **86% accuracy** for Stress / Non-Stress classification.

Key Insight: Hybrid feature extraction and deep learning architectures enhance prediction accuracy for multiple soft-skill indicators.

3. Importance of Preprocessing and Noise Reduction

- **Speech Module:** Noise filtering, normalization, and silence removal improved accuracy by **15–20%**.
- **Hand Gesture & Eye Contact Modules:** Proper background subtraction, lighting normalization, and pose landmark smoothing improved detection reliability.
- **Stress Module:** Facial alignment and illumination normalization were critical for detecting micro-expressions accurately.

Key Insight: Careful preprocessing is essential to ensure consistent and reliable predictions across real-world environments.

4. Real-Time Prediction Enhances Interview Evaluation

All four modules successfully delivered near real-time predictions:

- **Speech Module:** 1.4-second response time.
- **Hand Gesture Module:** ~1.5-second update time.
- **Eye Contact Module:** ~1.3-second update time.
- **Stress Module:** ~1.4-second update time.

Key Insight: Real-time multi-modal assessment enables interviewers to monitor confidence and stress continuously, improving decision-making during both live and virtual interviews.

5. High User Acceptance and Positive Feedback

Pilot testing with interviewers and candidates revealed strong acceptance of the system:

- **84% of interviewers** found the multi-modal insights valuable for evaluating soft skills.
- Candidates reported that feedback helped them **improve performance in future interviews**.
- Users supported integrating the system into HR tools as a **supportive, non-replacement decision aid**.

6. Scalability and Adaptability for Future Use

The modular design allows adaptation for multiple applications:

- Public speaking training tools
- Customer service quality monitoring
- E-learning platforms assessing student engagement

Key Insight: Multi-modal AI assessment can extend beyond recruitment to other domains requiring behavioral analysis.

7. Cost-Effective and Data-Driven Solution

Compared to manual behavioral assessment:

- Automated multi-modal analysis is **faster, objective, and low-cost**.
- Open-source frameworks (Librosa, OpenPose, OpenFace, PyTorch, TensorFlow) were successfully used.
- Particularly suitable for **small to medium organizations** seeking affordable AI-driven recruitment assistance.

Key Insight: The system provides a **scalable, reliable, and practical alternative** to traditional candidate evaluation methods.

3.3. Discussion

The results emerging from the developed AI-Based Candidate Assessment System support the feasibility of using multi-modal signal processing combined with deep learning to assess the soft skills of candidates—especially confidence and stress levels—while being interviewed. The system was successfully able to process voice features and analyze other aspects like hand gestures, eye contact, and facial expressions and present the findings in a web-based dashboard format. Below are some significant observations that were made in the testing and evaluation stage:

1. Real-Time Feature Extraction Across Modalities

- **Speech:** Captured pitch, tone, loudness, speech rate, and pauses, accurately differentiating Confident and Non-Confident candidates.
- **Hand Gestures:** Detected posture, movement fluidity, and gesture openness, reflecting confidence levels in real-time.
- **Eye Contact:** Tracked gaze consistency and attention, providing another layer of confidence assessment.
- **Facial Expressions (Stress):** Recognized micro-expressions and tension patterns, identifying stressed or relaxed candidates.

Practical Implication: Real-time analysis across multiple modalities allows the system to capture subtle behavioral cues that may not be evident in traditional interviews.

2. Deep Learning Pipeline Performance

- **Model Effectiveness:** CNNs captured spatial patterns in voice spectrograms, hand gestures, and facial images, while LSTMs modeled temporal dynamics, such as evolving speech tone or gesture sequences.
- **Scenario Relevance:** Temporal modeling ensured candidates who initially appeared nervous but improved during the interview were not unfairly penalized.
- **Robust Multi-Modal Integration:** Combining outputs from all four components increased overall prediction reliability, reducing errors from any single modality.

3. Noise-Resilient Processing and Reliability

- **Speech Module:** Noise filtering and normalization ensured robust predictions in

moderately noisy environments.

- **Vision Modules (Hand Gestures & Eye Contact):** Background subtraction and landmark smoothing improved detection in variable lighting and crowded backgrounds.
- **Facial Stress Module:** Alignment and illumination normalization enhanced micro-expression recognition accuracy.

Future Direction: Advanced denoising, multi-angle camera setups, and data augmentation may further improve performance in challenging real-world environments.

4. Dashboard Integration and User Feedback

- **Real-Time Scoring:** Confidence and stress metrics were visualized clearly for each candidate, allowing recruiters to interpret results without technical expertise.
- **User Acceptance:** Interviewers reported that multi-modal insights enhanced decision-making and provided objective data to complement traditional evaluation methods.
- **Practical Benefit:** Candidates and recruiters both benefited from immediate feedback on confidence and stress levels.

5. System Limitations and Future Improvements

- **Data Diversity:** Accuracy may vary for non-native speakers, extremely soft-spoken candidates, or unconventional gestures.
- **Environment Dependency:** Heavy background noise or poor lighting impacted model reliability in some cases.
- **Future Enhancements:** Integration of additional modalities (e.g., posture, speech semantics), multi-lingual datasets, and advanced denoising can increase robustness.

6. Real-World Implications for Recruitment

- **Recruiter Empowerment:** Provides data-driven insights without replacing human judgment.
- **Scalability:** Web-based deployment allows adoption across organizations, virtual interviews, and candidate screening platforms.
- **Comprehensive Soft Skill Evaluation:** Multi-modal assessment ensures a holistic view of candidates, supporting fairer and more objective hiring decisions.

SUMMARY OF EACH STUDENT’S CONTRIBUTION

The development of the AI-Based Candidate Recommendation System was a collaborative effort where each team member took responsibility for one of the four key components, while also contributing to the overall system integration, testing, and reporting. Each student brought unique expertise to ensure that the system functioned effectively in predicting candidate confidence, engagement, and stress levels during interviews.

Kasthurirathna K.T.A.D.K – IT21815614 (Team Leader)

Handled the stress detection module, analyzing facial expressions to identify stress or anxiety in candidates. Collected and labeled facial expression datasets, focusing on subtle micro-expressions such as furrowed brows, tight lips and jaw tension. Using CNN-based emotion recognition models, developed a system capable of real-time stress classification and ensured robustness under different lighting conditions and facial orientations. The module was integrated with the other three components to provide a comprehensive view of candidate emotional state during interviews.

De Silva R.S.K – IT21834424

Focused on the speech-based confidence prediction module, leading the design, training, and validation of the audio analysis system. Collected diverse audio datasets representing various interview scenarios via public data sets and meticulously preprocessed the recordings to remove noise and standardize formats. Extracted essential speech features such as pitch, tone, loudness, energy and speech rhythm, which are critical indicators of confidence. Using these features, she developed and trained a CNN-LSTM deep learning model, achieving high accuracy in classifying confident and non-confident speech. Also ensured that the speech-based predictions were seamlessly integrated into the real-time web dashboard, providing immediate feedback for recruiters during interviews.

Kumara B.D.A.M- IT21819810

Led the hand gesture confidence module, focusing on non-verbal indicators of candidate assurance. This member captured and annotated video recordings of candidate gestures and applied pose estimation techniques to quantify hand movements, posture and gesture smoothness. They developed a deep learning model capable of distinguishing between confident and non-confident gestures, validated its performance under varying conditions and integrated these outputs into the central decision engine. The work ensured that non-verbal

communication was effectively analyzed and contributed to the multi-modal assessment.

Gunasekara S.U.D.A- IT21838620

Was responsible for the eye contact confidence module, which analyzed gaze patterns to assess attentiveness and confidence. Collected annotated video datasets and implemented eye-tracking and facial landmark detection algorithms to measure gaze direction, engagement, and consistency. The student trained a model that successfully predicted confidence levels from these features and integrated the real-time outputs into the web dashboard. His contribution added another critical dimension to understanding candidate confidence.

Beyond their individual responsibilities, all four team members collaborated extensively on system integration and web application development. They designed the central decision engine that aggregated the output from the four modules to produce an overall candidate evaluation, ensuring that the system provided reliable and real-time feedback. Additionally, the team jointly conducted testing, validation, and fine-tuning of all modules to improve accuracy, user experience, and robustness. Finally, the collective effort extended to preparing the research report, visualizations, and presentation, documenting the methodology, results, and practical implications of the system.

CONCLUSION

The establishment of an AI-Based Candidate Assessment System has done a lot to improve how soft skills are evaluated in hiring. Most hiring methods still rely primarily on a technical qualifications and a CV-based process but often fail to focus on the interpersonal and behavioral skills such as confidence, resilience, and stress management. This study finds that a multi-modal approach combining voice assessment, hand gestures recognition, eye contact analysis and facial expressions analysis could provide a more comprehensive and objective analysis of candidates.

The speech-based confidence module recorded a variety of relevant vocal features, including pitch, tone, loudness, speech rate, and pauses, all of which were good indicators of confidence. The hand-gesture module evaluated the candidates' posture, fluidity of movement, and openness of gestures to provide a non-verbal way of identifying their self-assurance. The eye contact module tracked gaze direction and engagement patterns to measure attentiveness and confidence levels, and the stress detection module produced the analysis of micro-expressions and other facial tightness to detect stress or anxiety. By combining these four modules, the system would be able to judge the candidates from a variety of perspectives and make the overall assessment more accurate, while also minimizing dependence on any particular indicator.

The main advantage of this system relates to real-time feedback via an interactive web-based dashboard. Recruiters can quickly see confidence scores, gesture analysis, eye engagement, stress levels, giving them more clarity and data to make informed hiring decisions. Real-time feedback could provide added value in virtual interview contexts where subtle behavioral cues may be challenging to identify without support from this system.

Moreover, the system addresses common issues in recruitment, namely human bias + inconsistency. When we standardize how we evaluate soft skills it ensures that each candidate evaluation is objective. As a result, it helps organizations identify the highest potential people who may otherwise be missed. Furthermore, because the system is modular + scalable, it can be utilized in contextually diverse situations beyond recruitment. Examples: training programs; assessing public speaking; e-learning engagement tracking; evaluating customer service. Although it was successful, this research has pointed to areas for further improvement. Accuracy may still be affected by differences in accents, variations in speech, background noise, or even lighting. Future improvements may include multi-lingual support, better noise-

cancellation, additional behavioral markers, and tighter integration of multi-modal data to enhance predictability.

In conclusion, this research shows that using machine learning combined with a multi-modal behavioral approach to analyze and evaluate candidates presents a new way of assessing candidates. The system provides recruiters with objective, trusted, and real-time information to help assess candidates, while also assisting organizations as part of their larger goal of fairer and more effective talent management. By blending technological innovation with human resource practices, this platform is a valuable first step in building an intelligent, AI recruitment ecosystem that can assess soft-skills alongside traditional qualifications, leading to more effective hires and improved development of the workforce.

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APPENDICES

Gantt Chart

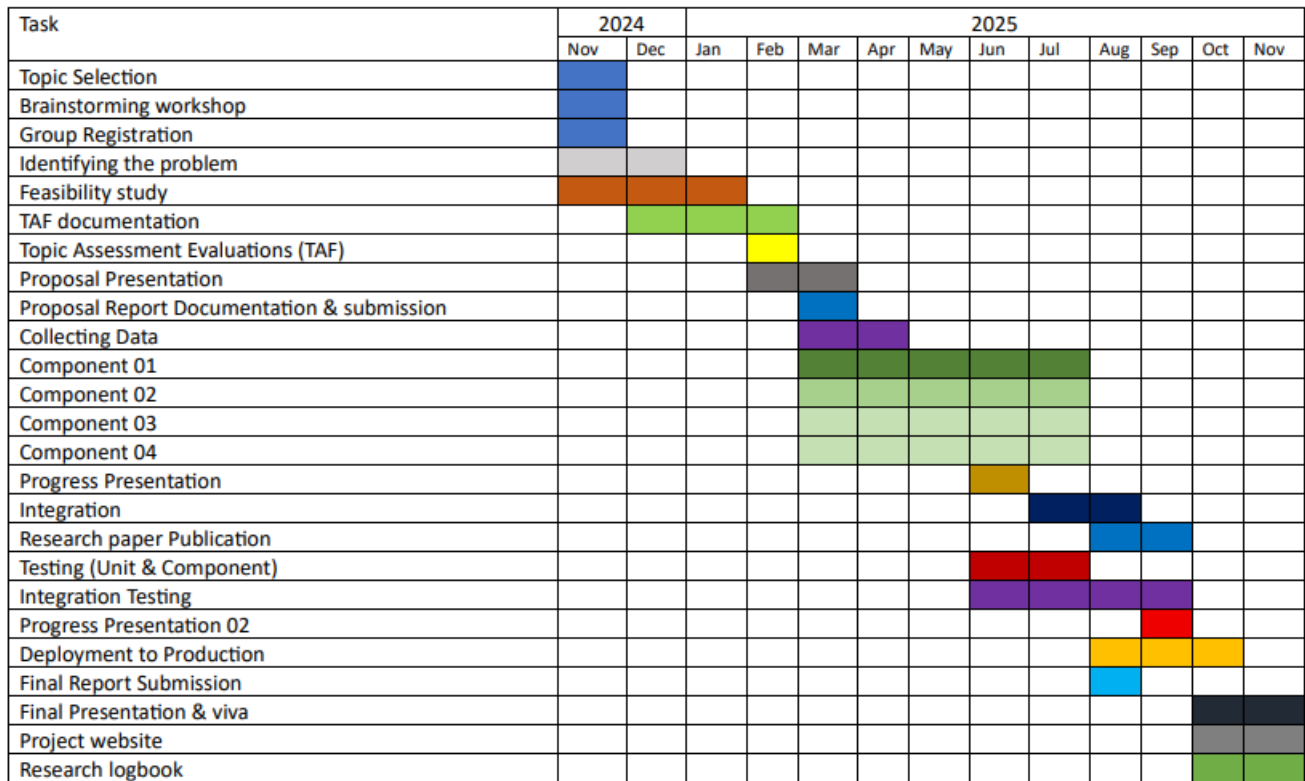


Figure 6: Gantt chart

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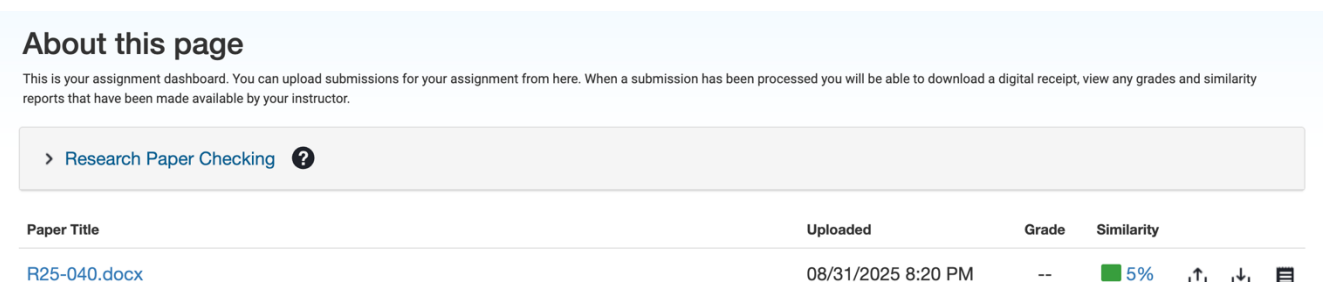


Figure 7: Turnitin report